I. FIELD AND PARTICLE NOTATION

The continuous entropy density field is denoted $S(x)$; the relativistic entropy density is

$$s_{\rm rel}(x) \equiv \frac{S(x)}{\gamma(x)}\,. \tag{1}$$

Each SPH particle a carries a conserved scalar s_a (entropy per particle). After smoothing, the SPH estimate of the entropy density at the particle site is

$$S(x_a) = \sum_b s_b\, W(x_a, x_b)\,. \tag{2}$$

The per-particle entropy satisfies

$$s_a \approx S(x_a)\,\mathrm{vol}_a\,, \tag{3}$$

where $\mathrm{vol}_a = L^2/N$ is the area per particle. This relation is *approximate*: $S(x_a)$ is itself a kernel average, so there is an $\mathcal{O}(h^2)$ smoothing error, and for a non-uniform lattice the exact equality never holds. The setup functions `set_ic!` and `setup_ic` in `shearwave.jl` initialise particles via $s_a = S_{\rm func}(x_a)\,\mathrm{vol}_a$, where $S_{\rm func}(x, y)$ is the user-supplied continuous entropy density.

II. ALGORITHM

We define the sph kernel $W(x_a, x_b) = F(\frac{1}{2}(x_a - x_b)^2)$ and the derivative

$$D_k(x_a, x_b) = \frac{\partial W(x_a, x_b)}{\partial x_a^k} \tag{4}$$

1. Given m_a^i and s_a compute $M^i(x_a), S(x_a)$.

$$M^i(x_a) = \sum_b m_b^i W(x_a, x_b) \tag{5}$$

$$S(x_a) = \sum_b s_b W(x_a, x_b) \tag{6}$$

2. Use the equation of state to find the thermodynamic conjugates, $(v/S)_i(x_a)$ and $-(\mathcal{F}/S)(x_a)$, from the extensive variables $M^i(x_a), S(x_a)$. Also compute the potential $\mathcal{E}(x_a)$ and (optionally) its legendre transform $T(x_a)$. See Sect. IV A 4 for details.

3. Compute a number of quantities

$$\dot{x}_{i,a} \equiv \sum_b s_b (v/S)_i(x_b) W(x_a, x_b) \tag{7}$$

$$\mathsf{T}_a = \mathcal{E}(x_a) - \sum_b s_b (\mathcal{F}/S)(x_b) W(x_a, x_b) \tag{8}$$

$$\mathsf{b}_{i,a} \equiv \frac{\dot{x}_{i,a}}{\mathsf{T}_a} \tag{9}$$

These are related to the derivatives of the Hamiltonian (see below)

4. The Hamiltonian is

$$\mathbb{H}(s_a, x_a, m_a) = \sum_a s_a \mathcal{E}(M(x_a), S(x_a)) \tag{10}$$

and we have (as advertised)

$$\Delta\mathbb{H}(s_a, x_a, m_a) = \sum_a \mathsf{T}_a \Delta s_a + \dot{x}_a \cdot \Delta m_a \tag{11}$$

The ideal equations are Hamilton's. Spefically we do the following: Update the positions

$$\Delta x_a^i = \Delta t\, \delta^{ik} \dot{x}_{k,a} \tag{12}$$

and then update the momentum due to the ideal terms

$$\Delta m_a^i = -\Delta t \sum_b \big[(v/S)_j(x_a) s_a m_b^j + (v/S)_j(x_b) s_b m_a^j \\ - (\mathcal{F}/S)(x_a) s_a s_b - (\mathcal{F}/S)(x_b) s_b s_a\big] \delta^{ik} D_k(x_a, x_b)\,. \tag{13}$$

5. Update the momentum and entropy due to viscous forces. Update the momentum by viscous forces

$$\Delta m^i_{\mathrm{visc},a} = \Delta t\, f^i_{\mathrm{visc}}(x_a) \tag{14}$$

$$\Delta s_a = -\,\mathsf{b}_a \cdot \Delta m_{\mathrm{visc},a} \tag{15}$$

The viscous force is assembled in two passes — the two-level scheme of (54), (55) and App. D. First compute the SPH strain at every particle,

$$\varepsilon_{ji}(x_a) = \frac{1}{S(x_a)} \sum_b s_b \left(\mathsf{b}_{i,b} - \mathsf{b}_{i,a}\right) D_j(x_a, x_b)\,, \tag{16}$$

then assemble the force from the stored strains,

$$f^i_{\mathrm{visc}}(x_a) = \sum_b F^i_{\mathrm{visc}}(x_a, x_b) = -\frac{\partial K}{\partial \mathsf{b}_{i,a}}\,, \tag{17}$$

$$F^i_{\mathrm{visc}}(x_a, x_b) = s_a s_b\, D_j(x_a, x_b) \left(\frac{T_{\mathrm{rel}} \kappa^{ijlm}}{S^2}\bigg|_a \varepsilon_{lm}(x_a) + \frac{T_{\mathrm{rel}} \kappa^{ijlm}}{S^2}\bigg|_b \varepsilon_{lm}(x_b) \right). \tag{18}$$

The total rate of entropy production is

$$\frac{d\mathbb{S}}{dt} \;=\; 2K \;=\; \sum_a \frac{s_a}{S(x_a)}\, \varepsilon_{ji}(x_a)\, T_{\mathrm{rel}} \kappa^{ijlm}\big|_a\, \varepsilon_{lm}(x_a) \;\geq\; 0\,, \tag{19}$$

with K the dissipative function (55).

III. STOCHASTIC UPDATES

First we generate the noise for each pair

$$\left\langle \xi^{ij} \xi^{lm} \right\rangle = 2 T_{\mathrm{rel}} \kappa^{ijlm}(v) \delta(t - t') \delta^2(\mathbf{r} - \mathbf{r}') \tag{20}$$

IV. HAMILTONIAN EVOLUTION

Starting point

$$\boxed{\mathbb{H}(s_a, x_a, m_a) = \sum_a s_a \mathcal{E}(\mathcal{M}(x_a), V_s(x_a))} \tag{21}$$

The variables in the Hamiltonian are summarized below.

Thermo: We have

$$d\mathbb{E} = Td\mathbb{S} + vd\mathbb{M} - pd\mathbb{V} \tag{22}$$

We work with a volume where $\mathbb{S} = \mathbb{S}_0$ is constant. Then $S = \mathbb{S}_0/\mathbb{V}$. Divinding by $\mathbb{S}_0$ and definining $d\mathbb{V}/\mathbb{V} = -dS/S(x)$ one has

$$d\mathcal{E} = vd\mathcal{M} - pd(1/S) \tag{23}$$

where $\mathcal{E} = \mathbb{E}/\mathbb{S}$ and $\mathcal{M} = \mathbb{M}/\mathbb{S}$. We also note

$$d\mathcal{E} = \frac{v}{S} dM - \frac{\mathcal{F}}{S} dS \tag{24}$$

where the free energy per entropy is $\mathcal{F}$:

$$\mathcal{F} \equiv \mathcal{E} - T = vM/S - p/S \qquad \mathcal{F}/S \equiv vM/S^2 - p/S^2 \tag{25}$$

We also note that $T \equiv -1/\beta_0(x)$ is the related to the "usual" $T_{\rm rel}$:

$$T \equiv \frac{1}{\beta^0} \equiv \frac{T_{\rm rel}}{\gamma}. \tag{26}$$

Variables: The variables are

$$S(x) = \frac{1}{V(x)} \equiv \sum_a s_a W(x, x_a) \tag{27}$$

$$M(x) \equiv \sum_a m_a W(x, x_a) \tag{28}$$

$$\mathcal{M}(x) \equiv \frac{1}{S(x)} \sum_a m_a W(x, x_a) \tag{29}$$

$$m_a \equiv \mathfrak{m}_a s_a \tag{30}$$

The conjugate variables are functions of these, i.e.

$$v(x_a) = v(\mathcal{M}(x), 1/S(x)) \neq v(\mathcal{M}_a, 1/s_a) \tag{31}$$

EOM: We use

$$\boxed{\frac{dx_a}{dt} = \frac{\partial \mathbb{H}}{\partial m_a}} \tag{32}$$

$$= \sum_b s_b v(x_b) W(x_b, x_a)/S(x_b) \tag{33}$$

$$\equiv S(x_a) \underline{v/S}(x_a) \tag{34}$$

where we use the underline to denote an average:

$$\underline{Q}(x) \equiv \frac{1}{S(x)} \sum_b Q_b \, s_b \, W(x_b, x) \tag{35}$$

Thus

$$\dot{x}_b - v(x_b) \equiv S(x_b)\underline{v/S}(x_b) - v(x_b) \tag{36}$$

And

$$\mathcal{M}(x) = \underline{\mathfrak{m}}(x) = \underline{m/s}(x) \tag{37}$$

We also have

$$\boxed{\frac{dm_a}{dt} = -\frac{\partial \mathbb{H}}{\partial x_a} + f^i_{\text{visc}}(x_a)} \tag{38}$$

Computing the derivatives we have

$$\frac{dm^i_a}{dt} = -\sum_b s_a s_b \left[(v/S)^j(x_a)\Delta\mathcal{M}^j_b(x_a) + (v/S)^j(x_b)\Delta\mathcal{M}^j_a(x_b)\right] \delta^{ik} D_k(x_a, x_b)$$
$$- \sum_b s_a s_b \left[(p/S^2)(x_a) + (p/S^2)(x_b)\right] \delta^{ik} D_k(x_a, x_b) + f^i_{\text{visc}}(x_a) \tag{39}$$

Here

$$\Delta\mathcal{M}_b(x_a) = \mathfrak{m}_b - \mathcal{M}(x_a) \tag{40}$$

We also defined the derivative operator

$$D_k(x_a, x_b) \equiv \frac{\partial W(x_a, x_b)}{\partial x^k_a} \tag{41}$$

In retrospect it seems natural to write the force term as

$$\frac{dm^i_a}{dt} = -\sum_b \left[(v/S)^j(x_a) s_a m_b + (v/S)^j(x_b) s_b m_a\right] \delta^{ik} D_k(x_a, x_b)$$
$$+ \sum_b \left[(\mathcal{F}/S)(x_a) s_a s_b + (\mathcal{F}/S)(x_b) s_b s_a\right] \delta^{ik} D_k(x_a, x_b) + f^i_{\text{visc}}(x_a) \tag{42}$$

Proposed viscous forces: When viscous forces are included we include the last line which we write just like the pressure.

$$\boxed{f^i_{\text{visc}}(x_a) \equiv -\sum_b s_a s_b \left((\Pi^{ik}/S^2)(x_a) + (\Pi^{ik}/S^2)(x_b)\right) D_k(x_a, x_b)} \tag{43}$$

$$\equiv \sum_b F^i_{\text{visc}}(x_a, x_b) \tag{44}$$

Then net momentum transfer is zero

$$\sum f_{\text{visc}}(x_a) = 0 \tag{45}$$

Change in entropy: Let's see how the energy changes under viscous evolution. We use the change in the entropy written as

$$\Delta S_{\rm visc}(x_a) = \sum_b \Delta s_b W(x_a, x_b) \tag{46}$$

Starting from eq. (21) we find

$$\Delta \mathbb{H} = \sum_a \left[\Delta s_a \, \mathcal{E}(\mathcal{M}, V^s) - s_a \, (\mathcal{F}/S)_a \Delta S_{\rm visc}(x_a)\right] + \sum_a \frac{\partial \mathbb{H}}{\partial m_a} \Delta m_{\rm visc,a} \tag{47}$$

Then noting

$$\sum_a s_a \, Q(x_a) \, \Delta S(x_a) = \sum_{ab} \Delta s_b \, W(x_b, x_a) \, s_a Q(x_a) = \sum_b \Delta s_b \, S(x_b) \, \underline{Q}(x_b) \tag{48}$$

the free energy term can be written

$$\sum_a s_a (\mathcal{F}/S)_a \, \Delta S_{\rm visc}(x_a) = \sum_b \Delta s_b \, S(x_b) \, \underline{\mathcal{F}/S}(x_b) \tag{49}$$

So we have

$$\Delta \mathbb{H} = \sum_a \Delta s_a \, \mathsf{T}_a + \frac{dx_a}{dt} \Delta m_{\rm visc,a} = 0 \tag{50}$$

where here:

$$\boxed{\mathsf{T}_a = T(x_a) + \Big(\mathcal{F}(x_a) - S(x_a) \underline{\mathcal{F}/S}(x_a)\Big)} \tag{51}$$

So to have energy conservation we have

$$\boxed{\Delta s_a = -\frac{1}{\mathsf{T}_a} \frac{dx_a}{dt} \cdot \Delta t \, f_a} \tag{52}$$

where we said $\Delta m_{\rm vis,a} = \Delta t f_a$. Then

$$\boxed{\Delta \mathbb{S} = \sum_a \Delta s_a = -\sum_{ab} \frac{\dot{x}_a}{\mathsf{T}_a} \cdot \Delta t \, F_{ab} \; = -\sum_{ab} \left(\frac{\dot{x}_a}{\mathsf{T}_a} - \frac{\dot{x}_b}{\mathsf{T}_b}\right) \cdot \Delta t \, F_{\rm visc}(x_a, x_b)} \tag{53}$$

Entropy production, deterministic hydro, and the dissipative function

We start by postulating the dissipative function Define the SPH strain at particle a as

$$\varepsilon_{ji}(x_a) \;\equiv\; \frac{1}{S(x_a)} \sum_b s_b \left(\mathsf{b}_{i,b} - \mathsf{b}_{i,a}\right) D_j(x_a, x_b) \,, \tag{54}$$

which is an antisymmetric SPH gradient estimator for $\partial_j \beta_i$ at particle a. The discrete dissipative function is then

$$\boxed{K \;=\; \frac{1}{2} \sum_a \frac{s_a}{S(x_a)} \, \varepsilon_{ji}(x_a) \, T_{\rm rel} \kappa^{ijlm}\big|_a \, \varepsilon_{lm}(x_a) \,.} \tag{55}$$

Here the two factors of D_j live in *separate* neighbor sums (one for each index of ε), so there is no overlap. The volume element $s_a/S(x_a)$ is the correct SPH approximation to d^2x. We have defined

$$\mathsf{b}_i(a) \equiv \frac{\dot{x}_{a,i}}{\mathsf{T}_a} \tag{56}$$

As we will see the rate of entropy production is

$$\dot{\mathbb{S}} = 2K \tag{57}$$

We take

$$\boxed{f^i_{\rm visc}(x_a) = -\frac{\partial K}{\partial \mathsf{b}_{i,a}} = \sum_b F^i_{\rm visc}(x_a, x_b)} \tag{58}$$

where the pair of forces is

$$\boxed{F^i_{\rm visc}(x_a, x_b) = s_a s_b\, D_j(x_a, x_b)\, \left(\frac{T_{\rm rel}\kappa^{ijlm}}{S^2}\bigg|_a\, \varepsilon_{lm}(x_a) \;+\; \frac{T_{\rm rel}\kappa^{ijlm}}{S^2}\bigg|_b\, \varepsilon_{lm}(x_b) \right)} \tag{59}$$

We note that

$$F^i_{\rm visc}(x_a, x_b) = -F^i_{\rm visc}(x_b, x_a) \tag{60}$$

due to the antisymmetry of $D_i(a, b)$. This fits into the general mould where momentum is conserved since

$$\sum_a f^i_{\rm visc}(x_a) = \sum_{ab} F^i(x_a, x_b) = 0 \tag{61}$$

So the previously defined mean strains are now more explicit:

$$\left(\frac{\Pi^{ij}}{S^2}(x_a) + \frac{\Pi^{ij}}{S^2}(x_b) \right) = -\left(\frac{T_{\rm rel}\kappa^{ijlm}}{S^2}\bigg|_a\, \varepsilon_{lm}(x_a) \;+\; \frac{T_{\rm rel}\kappa^{ijlm}}{S^2}\bigg|_b\, \varepsilon_{lm}(x_b) \right) \tag{62}$$

with

$$\boxed{F^i_{\rm visc}(x_a, x_b) \equiv -s_a s_b \left((\Pi^{ik}/S^2)(x_a) + (\Pi^{ik}/S^2)(x_b) \right) D_k(x_a, x_b)} \tag{63}$$

The signs and magnitude are consistent with the continous expression

$$\Pi^{ij} = -T_{\rm rel}\kappa^{ijlm}\partial_l \beta_m \tag{64}$$

The net force on a particle

$$f^i_{\rm visc}(x_a) = -\sum_b s_a s_b D_j(x_a, x_b) \left(\frac{\Pi^{ij}}{S^2}(x_a) + \frac{\Pi^{ij}}{S^2}(x_b) \right) \tag{65}$$

This is compatible with the continous expression

$$\partial_t M^i = -\partial_j \Pi^{ij} \tag{66}$$

Finally and most importantly

$$\boxed{\dot{\mathbb{S}} = \sum_a -\mathsf{b}_a \cdot f_{\rm visc}(x_a) = \mathsf{b}_a \cdot \frac{\partial K}{\partial \mathsf{b}_a} = \; 2K \geq 0} \tag{67}$$

A. Some notes

1. Translational Invariance

We have charges

$$Q(x) = \sum_a q_a W(x, x_a) \tag{68}$$

We have

$$W(1,2) = F(\tfrac{1}{2}(\vec{x}_1 - \vec{x}_2)^2) = F(1,2) \tag{69}$$

$$\frac{\partial W(1,2)}{\partial \vec{1}} = F'(1,2)\,(\vec{x}_1 - \vec{x}_2) \equiv W'(1,2) \tag{70}$$

$$\frac{\partial W(1,2)}{\partial \vec{2}} = -\,F'(1,2)(\vec{x}_1 - \vec{x}_2) \equiv -W'(1,2) \tag{71}$$

So

$$\frac{\partial Q(x_b)}{\partial x_a} = \left(\sum_c q_c W'_{ac}\right)\delta_{ab} - q_a W'_{ba} \qquad \text{no-}a\text{-sum! } r \tag{72}$$

We note that translational invariance implies

$$\sum_a \frac{\partial Q(x_b)}{\partial x_a} = 0 \tag{73}$$

2. Derivatives with respect to x_a

We have the variables

$$Q^I = (M, S) \tag{74}$$

Then

$$\mathcal{E}_I \equiv \frac{\partial \mathcal{E}}{\partial u^I} = (\frac{v}{s}, -\mathcal{F}/S) \tag{75}$$

$$\frac{\partial \mathbb{H}}{\partial x_a} = \sum_b s_b \frac{\partial \mathcal{E}}{\partial x_a} \tag{76}$$

$$= \sum_b s_b \mathcal{E}_I(x_b) \frac{\partial Q^I(x_b)}{\partial x_a} \tag{77}$$

$$= \sum_b s_b \mathcal{E}_I(x_b) \left[(\sum_c q^I_c W'_{ac})\delta_{ba} - q^I_a W'_{ba} \right] \tag{78}$$

$$= \sum_c s_a \mathcal{E}_I(x_a) q^I_c W'_{ac} - \sum_b q^I_a W'_{ba} s_b \mathcal{E}_I(x_b) \tag{79}$$

$$= \sum_b \left[s_a \mathcal{E}_I(x_a)\; q^I_b + s_b \mathcal{E}_I(x_b)\; q^I_a \right] W'_{ab} \tag{80}$$

3. Evaluation of $\partial\mathbb{H}/\partial x_a$

For the specific case we have

$$\frac{\partial\mathbb{H}}{\partial x_a} = \sum_b \left[s_a \left((v/S)(x_a) m_b - (vM/S^2)(x_a) s_b \right) + s_b \left((v/S)(x_b) m_a - (vM/S^2)(x_b) s_a \right) \right] W'_{ab} + \text{pressure} \quad (81)$$

Defining

$$\Delta\mathcal{M}_b(x_a) \equiv \mathfrak{m}_b - \mathcal{M}(x_a) \quad (82)$$

We have

$$\frac{\partial\mathbb{H}}{\partial x_a} = \sum_b s_a s_b \left[v/S(x_a)\, \Delta\mathcal{M}_b(x_a) + v/S(x_b)\, \Delta\mathcal{M}_a(x_b) \right] W'_{ab} + \text{pressure} \quad (83)$$

The pressure term is

$$\text{pressure} = \sum_b \left[s_a (p/S^2)(x_a) s_b + s_b (p/S^2)(x_b) s_a \right] W'_{ab} \quad (84)$$

$$= \sum_b s_a s_b \left[(p/S^2)(x_a) + (p/S^2)(x_b) \right] W'_{ab} \quad (85)$$

So to summarize

$$\frac{\partial\mathbb{H}}{\partial x_a} = \sum_b s_a s_b \left[v/S(x_a)\, \Delta\mathcal{M}_b(x_a) + v/S(x_b)\, \Delta\mathcal{M}_a(x_b) + (p/S^2)(x_a) + (p/S^2)(x_b) \right] W'_{ab} \quad (86)$$

4. Solving the equation of state

We have the momentum $M^i(x) = (e+p)\gamma^2 v^i$ and the entropy

$$S = \frac{(e+p)\,\gamma}{T_{\rm rel}(e)}\,, \quad (87)$$

where $T_{\rm rel}(e)$ is the relativistic temperature determined by the equation of state. Given $S_{\rm in} \equiv S(x_a)$ and $M \equiv |\vec{M}(x_a)|$, we want to find e and hence the thermodynamic conjugates (v/S) and $-(\mathcal{F}/S)$:

$$\mathrm{d}\mathcal{E} = (v/S)_i\, \mathrm{d}M^i - (\mathcal{F}/S)\, \mathrm{d}S\,, \quad (88)$$

where $\mathcal{E}$ is the energy per entropy. The spatial index is suppressed below, writing $M^i = Mn^i$ and $(v/S)_i = (v/S)\, n_i$ with n^i a unit vector.

Step 1: velocity from a trial e. With $w = e + p$, the momentum is $M^i = w\gamma^2 v^i = w\gamma\, u^i$ with $u^i = \gamma v^i$. Writing $G \equiv w\gamma$, so that $M = G\, u$, and using $\gamma^2 = 1 + u^2$,

$$G^4 - w^2 G^2 - w^2 M^2 = 0\,, \qquad G = \sqrt{\frac{w^2 + \sqrt{w^4 + 4w^2M^2}}{2}}\,, \qquad \gamma = \frac{G}{w}\,, \qquad v = \frac{M\, w}{G^2}\,. \quad (89)$$

This is exact given e; no sub-iteration is required. (Taking $u = M/w$ instead, as an earlier version of these notes and of the code did, drops a factor γ: the energy $(e+p)\gamma^2 - p$ is then not conjugate to M, and the Hamiltonian velocity and temperature are wrong at $O(v^2)$.)
Step 2: Newton iteration on e. Define the residual

$$f(e) = \frac{G(e)}{T_{\rm rel}(e)} - S_{\rm in}\,, \tag{90}$$

where $G(e) = G(w(e), M)$ is evaluated via Step 1. For a conformal equation of state f rises monotonically from $-S_{\rm in}$ at $e \to 0$ to $+\infty$, so there is a root for every M. The Newton derivative follows from the thermodynamic identity

$$\frac{dT_{\rm rel}}{de} = \frac{c_s^2\, T_{\rm rel}}{e+p} \tag{91}$$

and $dw/de = 1 + c_s^2$:

$$f'(e) = \frac{1}{T_{\rm rel}}\left[(1+c_s^2)\frac{dG}{dw} - \frac{c_s^2\, G}{w}\right], \qquad \frac{dG}{dw} = \frac{1}{2G}\left[w + \frac{w^3 + 2wM^2}{\sqrt{w^4 + 4w^2M^2}}\right]. \tag{92}$$

The Newton update is $e \leftarrow e - f(e)/f'(e)$.
Post-convergence expressions. With $S \equiv S_{\rm in}$:

$$p = p(e)\,, \tag{93}$$
$$\frac{v}{S} = \frac{v}{S}\,, \tag{94}$$
$$\frac{\mathcal{F}}{S} = \frac{v}{S}\frac{M}{S} - \frac{p}{S^2}\,, \tag{95}$$
$$\mathcal{E} = \frac{(e+p)\gamma^2 - p}{S}\,, \tag{96}$$
$$T = \frac{e+p}{S}\,. \tag{97}$$

V. DISSIPATIVE FUNCTION

We use the Hamiltonian formalism of DZYALOSHINSKII AND VOLOVICK section 2:

$$T\left(\partial_t S(x) + \{S(x), H\}\right) = T\partial_\ell\left(\frac{\partial R}{\partial \nabla_\ell T}\right) + TK \tag{98}$$

$$v^k\left(\partial_t M_k(x) + \{M_k(x), H\}\right) = v^k\partial_\ell\left(\frac{\partial R}{\partial \nabla_\ell v^k}\right) \tag{99}$$

Here $R = K = 0$ corresponds to ideal hydro. K is rate of entropy production. As we will see, $R = TK/2$, and R is the dissipative function. In the density frame we know that

$$K = T_{\rm rel}\kappa^{ijlm}\partial_i\beta_j\partial_l\beta_m \tag{100}$$

One of the goals of this section is to relate K to the density frame entropy current.

To see the relation between R and K we have

$$\partial_t E + \partial_k \mathcal{T}^{0k} = T\partial_\ell \left(\frac{\partial R}{\partial \nabla_\ell T} \right) + v^k \partial_\ell \left(\frac{\partial R}{\partial_\ell v^k} \right) + TK \tag{101}$$

$$= \partial_\ell \left(T \frac{\partial R}{\partial \nabla_\ell T} + v^k \frac{\partial R}{\partial \nabla_\ell v_k} \right) - (\nabla_\ell T) \left(\frac{\partial R}{\partial \nabla_\ell T} \right) - (\nabla_\ell v^k) \left(\frac{\partial R}{\partial \nabla_\ell v^k} \right) + TK \tag{102}$$

Demanding that the rhs is zero up to a total divergence, we find the relation between R and K. In general the dissipative functional involves

$$2R = TK = T T_{\rm rel} \kappa^{ijlm} \partial_i \beta_j \partial_\ell \beta_m \tag{103}$$

where

$$\partial_{(i}\beta_{j)} = \frac{1}{T} \left(-\frac{v_{(i}\partial_{j)} T}{T} + \partial_{(i} v_{j)} \right) \tag{104}$$

Then, the dissipative flux to the energy

$$\partial_\ell \left(T \frac{\partial R}{\partial \nabla_\ell T} + v^k \frac{\partial R}{\partial \nabla_\ell v_k} \right) = 0 \tag{105}$$

This is the density-frame condition

The strain according to us is

$$\Pi^{ij} = -T_{\rm rel} \kappa^{ijlm} \partial_{(\ell} \beta_{m)} \tag{106}$$

which does indeed work:

$$\Pi^{ij}_R = -\frac{\partial R}{\partial \nabla_\ell v_k} = -T_{\rm rel} \kappa^{ijlm} \partial_{(\ell} \beta_{m)} \, . \tag{107}$$

Then we have according to us:

$$\partial_t S + \partial_i (S v^i) = \partial_j (\beta_i \Pi^{ij}) + \partial \beta \cdot T_{\rm rel} \kappa \cdot \partial \beta \tag{108}$$

We identify the entropy flux from the dissipative function

$$\frac{\partial R}{\partial (\nabla_j T)} = -v_i \kappa^{ijlm} \partial_l \beta_m = \beta_i \Pi^{ij} \tag{109}$$

which is in agreement.

We see that the entropy conservation law can be written

$$\partial_t S + \partial_i (S v^i) = \beta_i \, \partial_j \Pi^{ij} \tag{110}$$

Dividing by N and using the conservation law

$$\partial_t N + \partial_i N v^i = 0 \tag{111}$$

$$D_t (S/N) = \frac{1}{N(x)} \beta_i \partial_j \Pi^{ij} \tag{112}$$

One can take $N = S^{(0)}(t, x)$ the initial entropy.

A. The κ^{ijlm} tensor

We use the $d = 3$ formula for κ^{ijlm} throughout, even though the SPH simulation is restricted to 2D motion in the (x, y) plane. This choice is physically natural: the fluid is genuinely three-dimensional and the simulation is a 2D cross-section. The $d = 3$ formula gives the standard coefficient $\kappa^{xxxx}|_{v=0} = 4\eta/3$ in the non-relativistic limit.

The tensor splits into shear and bulk parts,

$$T_{\rm rel}\kappa^{ijlm} = \kappa^{ijlm}_{\rm shear}(v)\,\frac{T_{\rm rel}\,\eta}{(1-c_s^2v^2)^2} + \kappa^{ijlm}_{\rm bulk}(v)\,\frac{T_{\rm rel}\,\zeta}{(1-c_s^2v^2)^2}\,, \tag{113}$$

where $i, j, l, m \in \{x, y\}$ are purely spatial indices and $v^2 = v_x^2 + v_y^2$. Define the spatial projector

$$L^{ij} = \delta^{ij} - v^i v^j \tag{114}$$

and the $d = 3$ abbreviations

$$\langle PL\rangle = 1 - c_s^2 v^2\,, \tag{115}$$

$$(LPL)^{ij} = -\frac{c_s^2\,v^i v^j}{\gamma^2} + \frac{L^{ij}}{3}\,, \tag{116}$$

$$\langle PLPL\rangle = \frac{2}{3}(c_s^2v^2)^2 + \frac{1}{3}(1-c_s^2v^2)^2\,. \tag{117}$$

Shear part:

$$\kappa^{ijlm}_{\rm shear} = 2\langle PL\rangle^2\left[L^{(il}L^{j)m} - \frac{L^{ij}(LPL)^{lm}}{\langle PL\rangle} - \frac{L^{lm}(LPL)^{ij}}{\langle PL\rangle} + \frac{\langle PLPL\rangle}{\langle PL\rangle^2}\,L^{ij}L^{lm}\right] \tag{118}$$

where $L^{(il}L^{j)m} = \frac{1}{2}(L^{il}L^{jm} + L^{im}L^{jl})$.

Bulk part:

$$\kappa^{ijlm}_{\rm bulk} = L^{ij}\,L^{lm} \tag{119}$$

For a conformal EOS, Weyl invariance requires $\zeta = 0$, so the bulk term vanishes.

Checks at $v = 0$:

$$T_{\rm rel}\kappa^{xxxx}\big|_{v=0} = \tfrac{4}{3}\,T_{\rm rel}\eta\,, \tag{120}$$

$$T_{\rm rel}\kappa^{xxyy}\big|_{v=0} = -\tfrac{2}{3}\,T_{\rm rel}\eta\,, \tag{121}$$

$$T_{\rm rel}\kappa^{xyxy}\big|_{v=0} = T_{\rm rel}\eta\,. \tag{122}$$

1D check ($v_y = 0$, speed $v = v_x$):

$$T_{\rm rel}\kappa^{xxxx}(v_x, 0) = \frac{4}{3}\,\frac{T_{\rm rel}\,\eta}{\gamma^4\,(1-c_s^2v_x^2)^2} \tag{123}$$

which agrees with the main-text result for κ^{xxxx}.

Symmetries:

$$\kappa^{ijlm} = \kappa^{jilm} = \kappa^{ijml} = \kappa^{lmij} \tag{124}$$

The tensor has only 6 independent components for 2D indices: κ^{xxxx}, κ^{xxxy}, κ^{xxyy}, κ^{xyxy}, κ^{xyyy}, κ^{yyyy}.

Appendix A: Implementation notes for the Julia SPH code

Field and particle notation

- s_a — entropy of particle a (extensive, units: entropy)
- m_a^i — momentum of particle a (extensive, units: momentum)
- $S(x) = \sum_a s_a\, W(x, x_a)$ — entropy density field (units: entropy/area, since W has units 1/area and $\int d^2x\, W = 1$)
- $M^i(x) = \sum_a m_a^i\, W(x, x_a)$ — momentum density field; satisfies $\int d^2x\, M^i(x) = \sum_a m_a^i$
- $\texttt{vol} = \Delta x^2$ — particle volume (area per particle on the initialization lattice); appears *only* in initialization to convert a density to a per-particle quantity, e.g. $s_a = s_{\text{density}} \times \texttt{vol}$. It does *not* appear in any smoothing sum.
- $s_{\text{rel}}(e) \equiv (e + p)/T_{\text{rel}}$ — relativistic entropy density (a thermodynamic field, not a per-particle quantity). *Note:* $s_{\text{rel}}(e) \neq s_a$. The approximate relation between them is $s_a \approx s_{\text{rel}}(e)\, \gamma\, \texttt{vol}$, since the per-particle entropy is the entropy density times the Lorentz-contracted volume element. The specific shear viscosity η/s appearing in the transport coefficients is properly η/s_{rel}; in Julia this is `eos.η_by_s` and the physical viscosity is $\eta = (\eta/s_{\text{rel}})\, s_{\text{rel}}(e)$.

Key definitions used in the dissipative step

a. The vector `b` *(`b_vec` in Julia).* The dissipative function K and the pair force $F^i_{\text{visc}}(x_a, x_b)$ involve the 2-vector $\mathsf{b}_a \equiv \dot{x}_a/\mathsf{T}_a$, where

$$\dot{x}_{i,a} = \sum_b s_b\, (v/S)_i(x_b)\, W(x_a, x_b) \quad \text{(kernel-smoothed velocity)}, \tag{A1}$$

$$\mathsf{T}_a = \mathcal{E}(x_a) - \sum_b s_b\, (\mathcal{F}/S)(x_b)\, W(x_a, x_b)\,. \tag{A2}$$

Here s_b is the per-particle entropy (not a density); the sum $\sum_b s_b W$ reproduces the entropy density $S(x_a)$ because $s_b = s_{\text{density}}\, \texttt{vol}$ was set at initialization. Concretely,

$$\texttt{b_vec}[a] = \dot{x}_a \;/\; \mathsf{T}_a\,. \tag{A3}$$

b. Hamiltonian structure behind $\dot{x}_a$ and T_a. Both quantities are Hamilton's equations for the discrete Hamiltonian

$$\mathbb{H}(s_a, x_a, m_a) = \sum_a s_a\, \mathcal{E}(M(x_a),\, S(x_a))\,, \tag{A4}$$

where $M(x_a) = \texttt{pa.M_smooth}$ and $S(x_a) = \texttt{pa.S_smooth}$ are the kernel-smoothed momentum and entropy densities. The energy per entropy $\mathcal{E}$ is evaluated as a function of the *smoothed* fields by `solve_eos!`:

$$\mathcal{E}(M_{\text{smooth}},\, S_{\text{smooth}}) \;=\; \texttt{pa.H_over_S} \;=\; \frac{(e+p)\gamma^2 - p}{S_{\text{smooth}}}\,. \tag{A5}$$

Its partial derivatives give the thermodynamic conjugates that `solve_eos!` stores:

$$\frac{\partial \mathcal{E}}{\partial M^i_{\rm smooth}} = \frac{v^i}{S} = \texttt{pa.v_over_S}[i]\,, \qquad \frac{\partial \mathcal{E}}{\partial S_{\rm smooth}} = -\mathcal{F}/S = -\texttt{pa.F_over_S}\,. \tag{A6}$$

Hamilton's equations then reproduce eqs. (A1)–(A2) exactly:

$$\dot{x}_{i,a} = \frac{\partial \mathbb{H}}{\partial m^i_a} = \sum_b s_b\, \texttt{pa_b.v_over_S}[i]\, W(x_a, x_b), \tag{A7}$$

$$\mathsf{T}_a = \frac{\partial \mathbb{H}}{\partial s_a} = \texttt{pa.H_over_S} - \sum_b s_b\, \texttt{pa_b.F_over_S}\, W(x_a, x_b)\,. \tag{A8}$$

Key point: `pa.H_over_S` is *not* the energy per entropy at the bare particle position; it is $\mathcal{E}$ evaluated at the *smoothed* state $(M_{\rm smooth}, S_{\rm smooth})$. Likewise, `pa.v_over_S` is $\partial\mathcal{E}/\partial M_{\rm smooth}$, computed inside `solve_eos!` via the thermodynamic identity $d\mathcal{E} = (v/S)\, dM - (\mathcal{F}/S)\, dS$.

c. Gotcha 1: $\mathsf{b} \neq v/T$. In relativistic kinetic theory one defines a thermodynamic inverse-temperature 4-vector $\beta^\mu = u^\mu/T_{\rm rel}$. Its spatial part is

$$\beta_i = \frac{u_i}{T_{\rm rel}} = \frac{\gamma v_i}{T_{\rm rel}} = \frac{v_i}{T_{\rm rel}/\gamma} = \frac{v_i}{\texttt{pa.T}}\,. \tag{A9}$$

This is *not* the same as b_i above. The two coincide only when (i) the SPH kernel average equals the pointwise value, i.e. the field is smooth on the scale h, *and* (ii) $\mathsf{T} = T$. In general b contains kernel-smoothing corrections through $\dot{x}$ (eq. (A1)) and through T (eq. (A2)). Using $v_i/\texttt{pa.T}$ in place of b_i in the pair force would violate energy conservation at $O(h^2)$.

d. Gotcha 2: $\dot{x}$ vs. the EOS velocity. After `smooth_to_particles!` and `update_thermodynamics!`, the particle holds $\texttt{pa.v_over_S} = v_i/S$ (the local thermodynamic 3-velocity divided by the local entropy density). The SPH-smoothed velocity $\dot{x}_a$ is *not* simply $\texttt{pa.v_over_S}[i] \times \texttt{pa.S_smooth}$; it is a kernel average over the neighbors' velocities (eq. (A1)), exactly as computed in `hamiltonian_velocity!`. In the function `compute_b_vec!` in `celllist.jl` both sums (for $\dot{x}$ and for $\sum_b s_b\,(F/S)(x_b)\,W$) are evaluated in the same neighbor loop.

e. Gotcha 3: $T_{\rm rel}\kappa^{ijlm}$ uses the local thermodynamic temperature. In the pair force

$$F^i_{\rm visc}(x_a, x_b) = \tfrac{1}{4} s_a s_b\, D_j(a,b) \left[\left.\frac{T_{\rm rel}\kappa^{ijlm}}{S^2}\right|_a + \left.\frac{T_{\rm rel}\kappa^{ijlm}}{S^2}\right|_b\right] D_l(a,b)\,(\mathsf{b}_{m,b} - \mathsf{b}_{m,a}), \tag{A10}$$

the factor $T_{\rm rel}\kappa^{ijlm}$ is returned by `kappa_tensor(pa, eos, Kappa3D())` and uses the *local* thermodynamic temperature $T_{\rm rel} = \texttt{pa.T} \times \texttt{pa.}\gamma$. This is correct and separate from the T appearing in b.

Appendix B: Sound waves test

For a constant background and small fluctuations we can neglect the differences between $E = (e+p)\gamma^2 - p \simeq e$ and $s_{\rm rel} \simeq S$, so that $\delta e(t,x) \simeq \frac{(e+p)}{S}\,\delta S \simeq T\,\delta S(t,x)$. Substituting $\delta e(t,x) = A\cos(kx - \omega t)$ and $M^x(t,x) = Ac_s\cos(kx-\omega t)$ into the linearized conservation laws,

$$\partial_t \delta e(t,x) + \partial_x M^x(t,x) = 0 \tag{B1}$$

$$\partial_t M^x(t,x) + c_s^2\, \partial_x \delta e(t,x) = 0 \tag{B2}$$

The solution has $\omega = c_s k$.

When dissipation is included the linearized equations become

$$\partial_t \delta e + \partial_x M^x = 0\,, \tag{B3}$$

$$\partial_t M^x + c_s^2\,\partial_x \delta e - \Gamma_s\,\partial_x^2 M^x = 0\,, \tag{B4}$$

where the sound-attenuation coefficient is

$$\Gamma_s \;=\; \frac{\frac{4}{3}\eta + \zeta}{e_0 + p_0} \;=\; \frac{\frac{4}{3}(\eta/s) + (\zeta/s)}{T_{\mathrm{rel},0}}\,. \tag{B5}$$

Inserting the plane-wave ansatz $\delta e, M^x \propto e^{i(kx-\omega t)}$ gives the exact dispersion relation

$$\omega^2 + i\,\Gamma_s k^2\,\omega - c_s^2 k^2 = 0\,. \tag{B6}$$

Solving exactly,

$$\omega \;=\; \pm\,\omega_R \;-\; i\,\frac{\Gamma_s k^2}{2}\,, \qquad \omega_R \;\equiv\; \frac{k}{2}\sqrt{4c_s^2 - \Gamma_s^2 k^2} \;=\; c_s k\sqrt{1 - \frac{\Gamma_s^2 k^2}{4c_s^2}}\,, \tag{B7}$$

valid in the underdamped regime $\Gamma_s k < 2c_s$. Both modes share the same decay rate $\Gamma_s k^2/2$; the phase velocity ω_R/k is shifted below c_s by a term of order $(\Gamma_s k/c_s)^2$. The decay timescale is $\tau_s = 2/(\Gamma_s k^2)$ regardless of whether the approximation $\Gamma_s k \ll c_s$ holds.

The right-moving eigenmode initialized at $t = 0$ with $\delta e(0, x) = A\cos(kx)$ evolves as

$$\delta e(t, x) = A\cos(kx - \omega_R t)\,e^{-\Gamma_s k^2 t/2}\,, \tag{B8}$$

$$M^x(t, x) = A\,e^{-\Gamma_s k^2 t/2}\left[\frac{\omega_R}{k}\cos(kx - \omega_R t) + \frac{\Gamma_s k}{2}\sin(kx - \omega_R t)\right]. \tag{B9}$$

The second line follows from the continuity equation $\partial_t \delta e + \partial_x M^x = 0$, which gives $M_0^x = \mathrm{Re}[(\omega/k)\,\delta e_0]$ with the *complex* $\omega/k = \omega_R/k - i\,\Gamma_s k/2$. A useful identity: $|\omega/k|^2 = (\omega_R/k)^2 + (\Gamma_s k/2)^2 = c_s^2$ (from the dispersion relation (B7)), so

$$M^x(t, x) = A\,c_s\,e^{-\Gamma_s k^2 t/2}\cos(kx - \omega_R t - \varphi)\,, \qquad \varphi = \arctan\!\left(\frac{\Gamma_s k^2}{2\,\omega_R}\right). \tag{B10}$$

The phase lag φ vanishes as $\Gamma_s \to 0$.

a. Initial condition. Using $\delta e \simeq T_{\mathrm{rel},0}\,\delta S$ the IC for the right-moving eigenmode is

$$\delta S(x) = \frac{A}{T_{\mathrm{rel},0}}\cos(kx)\,, \tag{B11}$$

$$M^x(x) = A\left[\frac{\omega_R}{k}\cos(kx) + \frac{\Gamma_s k}{2}\sin(kx)\right] = A\,c_s\cos(kx - \varphi)\,. \tag{B12}$$

For $\eta/s = \zeta/s = 0$ one has $\Gamma_s = 0$, $\omega_R = c_s k$, $\varphi = 0$, and the IC reduces to the one in `testsound.jl`. For the viscous test one sets $\eta/s > 0$ and uses the GENERIC integrator `generic_step!`.

Appendix C: Dissipative force per volume

We defined

$$M(x) = \sum m_a W(x, x_a) \tag{C1}$$

The force per volume is

$$\frac{\partial M(x)}{\partial t} = \sum_a \frac{dm_a}{dt} W(x, x_a) - m_a \dot{x}_a \partial_x W(x, x_a) \tag{C2}$$

For a small amplitude shear wave in a uniform background

$$M^y(x) = A \cos(kx) \tag{C3}$$

We have the following estimate:

$$\partial_t M^y(x) = -\partial_x \Pi^{yx} \simeq \partial_x (T_{\rm rel} \eta \, \partial_x \beta^y) \simeq \frac{\eta}{e+p} \partial_x^2 M^y = \frac{\eta/s}{T_0} \partial_x^2 M^y \tag{C4}$$

Appendix D: Two-level dissipative loop for the Julia SPH code

The problem with the original pair-based discretization

The dissipative function originally proposed in the main text is

$$K = \frac{1}{16} \sum_{ab} s_a s_b \, D_j(x_a, x_b)(\mathsf{b}_{i,b} - \mathsf{b}_{i,a}) \left[\left. \frac{T_{\rm rel} \kappa^{ijlm}}{S^2} \right|_a + \left. \frac{T_{\rm rel} \kappa^{ijlm}}{S^2} \right|_b \right] D_l(x_a, x_b)(\mathsf{b}_{m,b} - \mathsf{b}_{m,a}) \, . \tag{D1}$$

In the continuum limit this should approximate

$$K \; \longrightarrow \; \frac{1}{2} \int d^2x \; \partial_j \beta_i \, T_{\rm rel} \kappa^{ijlm} \, \partial_l \beta_m \, . \tag{D2}$$

The inconsistency is that $D_j(x_a, x_b)$ and $D_l(x_a, x_b)$ are both kernel gradients evaluated at the *same pair* (a, b). Their supports overlap completely, so the pair sum does not factorize into two independent gradient estimates at a point. In the continuum limit the product of two coincident kernel gradients produces a $\delta(0)$ divergence, and the normalization is wrong. Moreover, the pair contraction $D_j(a,b) D_l(a,b) \Delta\mathsf{b}_j \Delta\mathsf{b}_l$ captures only the *longitudinal* component of the strain along $\hat{r}_{ab}$; transverse components of $\partial_j \beta_i$ are missed at $\mathcal{O}(h^2)$.

The two-level remedy

The fix is to compute the strain tensor ε_{ji} at each particle site *first*, using a proper SPH gradient estimator, and then to build the force from the stored strains. This separates the two kernel gradients into independent neighbor sums, eliminating the overlap.

a. Discretize the continuum dissipative function. Define the SPH strain at particle a as

$$\varepsilon_{ji}(x_a) \;\equiv\; \frac{1}{S(x_a)} \sum_b s_b \left(\mathsf{b}_{i,b} - \mathsf{b}_{i,a}\right) D_j(x_a, x_b)\,, \tag{D3}$$

which is an antisymmetric SPH gradient estimator for $\partial_j \beta_i$ at particle a. The discrete dissipative function is then

$$K \;=\; \frac{1}{2} \sum_a \frac{s_a}{S(x_a)}\; \varepsilon_{ji}(x_a)\, T_{\rm rel} \kappa^{ijlm}\big|_a\, \varepsilon_{lm}(x_a)\,. \tag{D4}$$

Here the two factors of D_j live in *separate* neighbor sums (one for each index of ε), so there is no overlap. The volume element $s_a/S(x_a)$ is the correct SPH approximation to d^2x.

b. Viscous force from $\partial K/\partial \mathsf{b}_{i,c}$. Taking the derivative of K with respect to $\mathsf{b}_{i,c}$ and using $f^i_{\rm visc}(x_c) = -\partial K/\partial \mathsf{b}_{i,c}$ yields two terms, depending on whether c appears as the reference particle (the $a = c$ term) or as a neighbor (the $b = c$ term in the sum over b):

$$f^i_{\rm visc}(x_c) = \underbrace{+\frac{s_c}{S_c^2} \sum_b s_b\, D_j(c,b)\, T_{\rm rel}\kappa^{ijlm}\big|_c\, \varepsilon_{lm}(x_c)}_{\text{term 1: } c \text{ is reference particle}} \;-\; \underbrace{\sum_a \frac{s_a s_c}{S_a^2}\, D_j(a,c)\, T_{\rm rel}\kappa^{ijlm}\big|_a\, \varepsilon_{lm}(x_a)}_{\text{term 2: } c \text{ is neighbor of } a}\,. \tag{D5}$$

This can be written

$$f^i_{\rm visc}(x_c) = \sum_a s_c s_a\, D_j(c,a) \left(\left.\frac{T_{\rm rel}\kappa^{ijlm}}{S^2}\right|_c \varepsilon_{lm}(x_c) \;+\; \left.\frac{T_{\rm rel}\kappa^{ijlm}}{S^2}\right|_a \varepsilon_{lm}(x_a) \right) \tag{D6}$$

Finally relabelling we find

$$f^i_{\rm visc}(x_a) = \sum_b s_a s_b\, D_j(x_a, x_b) \left(\left.\frac{T_{\rm rel}\kappa^{ijlm}}{S^2}\right|_a \varepsilon_{lm}(x_a) \;+\; \left.\frac{T_{\rm rel}\kappa^{ijlm}}{S^2}\right|_b \varepsilon_{lm}(x_b) \right) \tag{D7}$$

Momentum conservation $\sum_c f^i_{\rm visc}(x_c) = 0$ follows from the antisymmetry of $D_j(x_a, x_b)$. We write this as in the main text

$$\left(\frac{\Pi^{ij}}{S^2}(x_a) + \frac{\Pi^{ij}}{S^2}(x_b)\right) = -\left(\left.\frac{T_{\rm rel}\kappa^{ijlm}}{S^2}\right|_a \varepsilon_{lm}(x_a) \;+\; \left.\frac{T_{\rm rel}\kappa^{ijlm}}{S^2}\right|_b \varepsilon_{lm}(x_b) \right) \tag{D8}$$

so

$$f^i_{\rm visc}(x_a) = -\sum_b s_a s_b\, D_j(x_a, x_b) \left(\frac{\Pi^{ij}}{S^2}(x_a) + \frac{\Pi^{ij}}{S^2}(x_b)\right) \tag{D9}$$

Entropy production $\dot S = 2K \geq 0$ is guaranteed because K is a positive-definite quadratic form in ε and κ^{ijlm} is positive definite.

c. Entropy update. Once $f^i_{\rm visc}(x_c)$ is known, the entropy is updated from energy conservation (eq. 50 of the main text):

$$\Delta s_a = -\mathsf{b}_a \cdot \Delta t\, f_{\rm visc,a}\,. \tag{D10}$$

Implementation in Julia — three passes

The two-level force requires two neighbor loops plus one particle loop. The total cost is approximately 3× the standard pair-force loop, but the continuum limit is correct.

d. Pass 1 — compute strain tensor at every particle. Loop over ALL neighbors $b \neq a$ (not just $b > a$, since $\varepsilon_{ji}(x_a)$ needs contributions from every neighbor):

```
for a in 1:N
    ε[a] = zero(SMatrix{2,2,Float64})
    for b in neighbors(a)            # all b ≠ a
        r_mic, r_mag = minimum_image(x[a]-x[b], L)
        r_mag ≥ 2h && continue
        D_ab  = wendland_c2_grad(r_mic, r_mag, h)
        Δb    = b_vec[b] - b_vec[a]
        ε[a] += s[b] * (Δb * D_ab^T) / S_smooth[a]   # outer product
    end
end
```

e. Pass 2 — compute viscous force from stored strains. Loop over pairs $b > a$ and apply Newton's third law:

```
for a in 1:N, b in neighbors(a) with b > a
    r_mic, r_mag = minimum_image(x[a]-x[b], L)
    r_mag ≥ 2h && continue
    D_ab   = wendland_c2_grad(r_mic, r_mag, h)
    κ_a    = kappa_tensor(particles[a], eos)     # T_rel κ^{ijlm} at a
    κ_b    = kappa_tensor(particles[b], eos)
    w_a    = s[a] * s[b] / S_smooth[a]^2
    w_b    = s[a] * s[b] / S_smooth[b]^2
    f_ab   = w_a * κ_a * ε[a] * D_ab
           - w_b * κ_b * ε[b] * D_ab
    f_visc[a] += f_ab
    f_visc[b] -= f_ab                           # Newton's third law
end
```

f. Pass 3 — entropy update. One loop over particles, no neighbor search required:

```
for a in 1:N
    Δs[a] = -dot(b_vec[a], dt * f_visc[a])
    s[a] += Δs[a]
end
```

Checks to run after implementation

1. **Momentum conservation:** $\sum_a f^i_{\text{visc}}(x_a) = 0$ to machine precision after Pass 2.
2. **Second law:** $\sum_a \Delta s_a \geq 0$ after Pass 3.

3. **Energy conservation:** $|\sum_a \mathsf{T}_a \, \Delta s_a + \dot{x}_a \cdot \Delta t \, f_{\mathrm{visc,a}}| < \epsilon_{\mathrm{tol}}$ (GENERIC degeneracy condition).

4. **Shear wave damping:** with $\eta > 0$, $\zeta = 0$, a transverse wave $M^y(x) = A\cos(kx)$ should damp at rate $\gamma_k = (\eta/s)\, k^2/T_0$ (Appendix D of main text).

Appendix E: Generating noise

We first generate a *unit-normalized* noise $\hat{\xi}^{ij}$ whose correlation function equals the bare kappa tensor,

$$\left\langle \hat{\xi}^{ij} \hat{\xi}^{lm} \right\rangle = \kappa^{ijlm} \,, \tag{E1}$$

with κ^{ijlm} the tensor built from $2\eta(\dots) + \zeta(\dots)$ below (no factor of T_{rel}). The physical density-frame noise is then the rescaling

$$\boxed{\xi^{ij} \;=\; \sqrt{2\, T_{\mathrm{rel}}}\, \hat{\xi}^{ij}} \tag{E2}$$

which satisfies the usual fluctuation–dissipation relation $\left\langle \xi^{ij}(x,t)\, \xi^{lm}(x',t') \right\rangle = 2\, T_{\mathrm{rel}}\, \kappa^{ijlm}(v)\, \delta(t-t')\, \delta^d(x-x')$, matching $\Pi^{ij} = -T_{\mathrm{rel}} \kappa^{ijlm} \partial_l \beta_m$. The construction of $\hat{\xi}^{ij}$ is done for every particle as follows:

1. Produce the Landau noise in the local rest frame $\xi^{\mu\nu}_{LRF}$ (this is a symmetric tensor purely spatial). Any component of this tensor with a zero index is zero

$$\xi^{0i}_{LRF} = \xi^{i0}_{LRF} = \xi^{00} = 0 \tag{E3}$$

   We need the covariance matrix of ξ^{ij}_{LRF} to satisfy

$$\left\langle \xi^{ij}_{LRF} \xi^{lm}_{LRF} \right\rangle = 2\eta \left[\delta^{(il}\delta^{j)m} - \frac{1}{d}\delta^{ij}\delta^{lm} \right] + \zeta \delta^{ij}\delta^{lm} \tag{E4}$$

   One way to do this is to pick one tensor

$$\xi^{ij}_L = \sqrt{d(d+2)\eta}\, \xi_1 \left(n^i n^j - \tfrac{1}{d}\delta^{ij} \right) + \sqrt{\zeta}\, \xi_2 \delta^{ij} \tag{E5}$$

   Here ξ_1 and ξ_2 are independent random numbers with variance one, and n^i is a unit vector drawn uniformly from the sphere ($d = 3$) or circle ($d = 2$). The relevant moments of the uniform measure are

$$\left\langle n^i n^j \right\rangle = \frac{1}{d}\, \delta^{ij} \tag{E6}$$

$$\left\langle n^i n^j n^l n^m \right\rangle = \frac{1}{d(d+2)} \left(\delta^{ij}\delta^{lm} + \delta^{il}\delta^{jm} + \delta^{im}\delta^{jl} \right) \tag{E7}$$

   One can then verify directly that the covariance is correct. Using the cross terms

$$\left\langle \left(n^i n^j - \tfrac{1}{d}\delta^{ij} \right) \left(n^l n^m - \tfrac{1}{d}\delta^{lm} \right) \right\rangle = \frac{\delta^{il}\delta^{jm} + \delta^{im}\delta^{jl}}{d(d+2)} - \frac{2}{d^2(d+2)}\, \delta^{ij}\delta^{lm} \tag{E8}$$

one finds

$$\left\langle \xi_L^{ij} \xi_L^{lm} \right\rangle = d(d+2)\eta \left[\frac{\delta^{il}\delta^{jm} + \delta^{im}\delta^{jl}}{d(d+2)} - \frac{2\,\delta^{ij}\delta^{lm}}{d^2(d+2)} \right] + \zeta\,\delta^{ij}\delta^{lm}$$
$$= 2\eta \left[\delta^{(il}\delta^{j)m} - \frac{1}{d}\delta^{ij}\delta^{lm} \right] + \zeta\,\delta^{ij}\delta^{lm} \tag{E9}$$

as required.

2. Boost to the lab frame. The fluid cell has four-velocity $u^\mu = (\gamma, \gamma v^i)$ in the lab frame. The Lorentz boost that takes the LRF rest vector $(1,0,0,0)$ to u^μ has matrix elements

$$\Lambda^0{}_0 = \gamma, \qquad \Lambda^0{}_j = \gamma\, v_j, \tag{E10}$$
$$\Lambda^i{}_0 = \gamma\, v^i, \qquad \Lambda^i{}_j = \delta^i_j + \frac{\gamma - 1}{v^2}\, v^i v_j, \tag{E11}$$

where $v^2 = v_i v^i$. The Landau-frame noise tensor in the lab frame is then

$$\xi_L^{\mu\nu} = \Lambda^\mu{}_\alpha\, \Lambda^\nu{}_\beta\, \xi_{LRF}^{\alpha\beta}\,. \tag{E12}$$

Since $\Delta^{\mu\nu} = g^{\mu\nu} + u^\mu u^\nu$ (with $g^{\mu\nu} = \mathrm{diag}(-1,1,1,1)$) is the projector onto the space orthogonal to u^μ, one can verify that

$$\left\langle \xi_L^{\mu\nu}\, \xi_L^{\alpha\beta} \right\rangle = 2\eta \left(\Delta^{(\mu\alpha}\Delta^{\nu)\beta} - \frac{1}{d}\Delta^{\mu\nu}\Delta^{\alpha\beta} \right) + \zeta\,\Delta^{\mu\nu}\Delta^{\alpha\beta}\,. \tag{E13}$$

3. Apply the frame-change matrix $\kappa^{ij}_{\mu\nu}$ to obtain the unit-normalized Density-Frame noise:

$$\hat{\xi}^{ij} = \kappa^{ij}_{\mu\nu}\, \xi_L^{\mu\nu}\,. \tag{E14}$$

The matrix $\kappa^{ij}_{\mu\nu}$ is built from the comoving-frame basis vectors and the equation of state. Define

$$e^i_\mu = \delta^i_\mu - v^i \delta^0_\mu\,, \tag{E15}$$
$$h^{ij} = e^i_\mu e^j_\nu \eta^{\mu\nu} = \delta^{ij} - v^i v^j\,. \tag{E16}$$

Here e^i_μ projects a lab-frame covariant index onto the comoving spatial directions, and h^{ij} is the induced spatial metric in the comoving frame. The transformation matrix is (theory.pdf eq. (43))

$$\kappa^{ij}_{\mu\nu} = e^i_{(\mu} e^j_{\nu)} - h^{ij}\, \frac{\partial \mathcal{P}}{\partial \mathcal{T}^{0\rho}} \delta^0_{(\mu} \delta^\rho_{\nu)}\,, \tag{E17}$$

where the pressure derivative (theory.pdf eq. (44)) is

$$\frac{\partial \mathcal{P}}{\partial \mathcal{T}^{0\rho}}\, \delta^0_\mu \delta^\rho_\nu = -\frac{c_s^2}{1 - v^2 c_s^2} \left[\frac{1}{\gamma^2} \Delta^0_\mu \Delta^0_\nu - u_\mu u_\nu \right]\,. \tag{E18}$$

Here $\Delta^{\mu\nu} = g^{\mu\nu} + u^\mu u^\nu$ is the projector orthogonal to u^μ, with $\Delta^0_\mu \equiv g^{0\nu}\Delta_{\nu\mu}$. It is also useful to introduce the scalar projector (theory.pdf eq. (45))

$$P_{\mu\nu} = -c_s^2\, u_\mu u_\nu + \frac{1}{d}\,\Delta_{\mu\nu}\,, \tag{E19}$$

in terms of which $\kappa^{ij}_{\mu\nu}$ decomposes into shear and bulk parts (theory.pdf eq. (46)):

$$\kappa^{ij}_{\mu\nu} = \underbrace{\left(e^i_{(\mu}e^j_{\nu)} - \frac{h^{ij}}{\langle Ph\rangle}(ePe)_{\mu\nu}\right)}_{\hat{\kappa}^{ij}_{\mu\nu}} + \underbrace{\frac{h^{ij}}{\langle Ph\rangle}\,P_{\mu\nu}}_{\bar{\kappa}^{ij}_{\mu\nu}}\,, \tag{E20}$$

where

$$(ePe)_{\mu\nu} = e^i_\mu P_{ij} e^j_\nu = -\frac{c_s^2}{\gamma^2}\,\Delta^0_\mu\Delta^0_\nu + \frac{1}{d}\,\Delta_{\mu\nu}\,, \tag{E21}$$

$$\langle Ph\rangle = P_{ij}\,h^{ji} = 1 - c_s^2 v^2\,. \tag{E22}$$

The covariance of the resulting unit-normalized Density-Frame noise then satisfies $\left\langle \hat{\xi}^{ij}\hat{\xi}^{lm}\right\rangle = \kappa^{ijlm}$ with (theory.pdf eq. (54))

$$\kappa^{ijlm} = \kappa^{ij}_{\mu\nu}\left[2\eta\big(\Delta^{(\mu\rho}\Delta^{\nu)\sigma} - \tfrac{1}{d}\Delta^{\mu\nu}\Delta^{\rho\sigma}\big) + \zeta\,\Delta^{\mu\nu}\Delta^{\rho\sigma}\right]\kappa^{lm}_{\rho\sigma}\,, \tag{E23}$$

and the physical noise $\xi^{ij} = \sqrt{2T_{\rm rel}}\,\hat{\xi}^{ij}$ of eq. (E2) then obeys the FDT normalization $\left\langle \xi^{ij}\xi^{lm}\right\rangle = 2T_{\rm rel}\kappa^{ijlm}$.

Appendix F: First stochastic steps

1. Physical Parameters

We need to decide on the parameters

- Shear Length

$$\ell \equiv \frac{\eta}{(e/c^2 + p)c_s} \tag{F1}$$

  we note that $sT = e + c^2 p$.

- Interparticle spacing

$$\ell_Q \equiv \frac{1}{(s_0)^{1/2}} \tag{F2}$$

- Speed of light c
- Temperature T_0

We can choose units where $c = T_0 = \ell_Q = 1$.

2. Defined and Derived Physical Parameters

Then we derive and define:

- We define the thermal momentum

$$p_0 \equiv \frac{T_0}{c} \tag{F3}$$

- The dimensionless shear viscosity is the mean free path in these units and reads

$$\hat{\eta} \equiv \ell/\ell_Q > 1 \tag{F4}$$

- The planks constant comes up in the definition of the entropy density

$$s_0 \equiv \frac{\kappa^2 p_0^2}{\hbar^2} \tag{F5}$$

and thus

$$\ell_Q = \frac{\hbar}{\kappa\, p_0} \tag{F6}$$

For a classical relativistic ideal gas with N^2 degrees of freedom in 2d we have

$$\kappa^2 \equiv \frac{N^2}{\pi} \tag{F7}$$

We find (using that $s_0 T_0 = e + c^2 p$) we have

$$\hat{\eta} = \left(\frac{\eta}{s\hbar}\right) \kappa \left(\frac{c}{c_s}\right) \tag{F8}$$

Numerically

$$\kappa = 3.4 \left(\frac{N^2}{37}\right)^{1/2} \tag{F9}$$

and for $c_s = c/2$

$$\hat{\eta} = 0.54 \left(\frac{4\pi\eta/s}{\hbar}\right) \left(\frac{\kappa}{3.4}\right) \tag{F10}$$

3. Basic description

We are applying hydro over a length L. There are viscous corrections

$$\text{classical} \sim \frac{\ell}{L} \tag{F11}$$

There are also $1/\sqrt{N}$ corrections

$$\text{noise} \sim \frac{1}{\sqrt{L^2 s}} = \frac{\ell_Q}{L} \tag{F12}$$

which in two dimensions scale similarly to the classical terms.

The ratio of these corrections is

$$\frac{\text{classical}}{\text{noise}} \sim \hat{\eta} \tag{F13}$$

4. Numerical Parameters

- h coarsening length
- a inter-sph-particle spacing

We want to choose h to be of order ℓ and a to be of order ℓ_Q or smaller. So we choose

$$a \sim \ell_Q \lesssim h \sim \ell \lesssim L \tag{F14}$$

To get started lets go crazy taking $\hat{\eta} \sim 10$

5. Relation to the code's $T_{\rm scale}$ normalization

In `sph.jl`, `ConformalEOS` parameterizes the temperature via

$$T_{\rm rel}(e) = T_{\rm scale}\, e^{c_s^2/(1+c_s^2)}\,. \tag{F15}$$

For the $d = 2$ fluid ($c_s^2 = 1/2$) this gives $e = (T_{\rm rel}/T_{\rm scale})^3$, and with $p = c_s^2 e$,

$$s = \frac{e+p}{T_{\rm rel}} = \frac{3}{2}\,\frac{T_{\rm rel}^2}{T_{\rm scale}^3}\,. \tag{F16}$$

Comparing to $s_0 = \kappa^2 p_0^2/\hbar^2 = (N^2/\pi)\,T_0^2$ (natural units $\hbar = c = 1$, $T_0 = T_{\rm rel}$) and matching the coefficients of T^2,

$$T_{\rm scale} = \left(\frac{3\pi}{2N^2}\right)^{1/3} = \frac{(3\pi/2)^{1/3}}{N^{2/3}}\,, \qquad \text{equivalently} \qquad N^2 = \frac{3\pi}{2\,T_{\rm scale}^3}\,. \tag{F17}$$

Using $\kappa^2 = N^2/\pi$ (line above) to eliminate N^2, this simplifies to

$$T_{\rm scale} = \left(\frac{3}{2\kappa^2}\right)^{1/3} = \frac{(3/2)^{1/3}}{\kappa^{2/3}}\,, \qquad \text{equivalently} \qquad \kappa^2 = \frac{3}{2\,T_{\rm scale}^3}\,, \tag{F18}$$

i.e. $T_{\rm scale}$ scales as $\kappa^{-2/3}$, with the π absorbed into κ leaving the clean coefficient $(3/2)^{1/3} \approx 1.145$. As a check, the numerical example above ($N^2 = 37 \Rightarrow \kappa \approx 3.4$) gives $T_{\rm scale} \approx 0.50$ — noticeably different from the arbitrary placeholder $T_{\rm scale} = 1.0$ currently used in `testeos2d.jl`, which only exercises the EOS-solver mechanics and was never meant to be physically normalized.

6. Checking equilibration

A first step is to choose and